Section	Description
Purpose	This workbook analyses synthetic laboratory results for 28 fictional patients, supplied as part of a data-analysis course. It contains no real patient data. It tries to answer the following four questions: 1-How do LDL levels compare by gender and age, within this sample of patients in the study? Who is most frequently above 130 mg/dL? 2-How do low hemoglobin levels differ by gender and age, within this sample of patients in the study? Who are most often below 11 g/dL (as a potential indicator of anemia)? 3-How do patients with ALT above 56 U/L compare with the remaining patients in age and gender composition? 4-Among patients with LDL (“bad cholesterol”) above 130 mg/dL, what proportion also have HDL (“good cholesterol”) below 40 mg/dL? How does this compare with patients whose LDL is within the supplied range?
Sample	The population sample of this analysis is 28 patients who have been tested for 13 values. This amounts to 364 results for all. The age of participants ranges between 25 and 74, with a median age of 50. Given the size of the sample, the best possibility to obtain equally sized group was to split patiens into two age groups **<50** and **50+**, with 14 patients each. Furthermore, there is an equal distribution of female and male patients, 14 for each.
How to use	The sheet "Sample" shows the calculations and distribution of the population sample for this study. "Main measures" contain 4 pivot tables addressing the 4 main questions of this project. Related to this "Charts" provide the visuals for the first three questions. Both "Main measures" and "Charts" have gender and age slicers which filter the connected tables and charts. Clear both to see the full sample. "Patient results" shows the main source of analysis and classifies each observation against supplied reference intervals on "dim_reference_intervals". "Patient Checker" allows to test the results of new patients and "dim_patients" show all the fictional patient demographic and contact details.
Limitations	The sample size is very small, so results should be interpreted only as exploratory. The gender composition is unequal within age groups; there are more younger women than men; and conversely, more older men than women, which has also an effect on the results of measures which act as proxies for certain conditions. The supplied thresholds on dim_reference_intervals, have provided the basis for evaluating the results; however, the HDL upper limit is still unresolved.
Main observations	· High LDL: 42.9% of patients (12/28) had high LDL. The proportion was higher in the 50+ group (78.6%) than in the under-50 group (7.1%). · Low haemoglobin: 17.9% of patients (5/28) had low haemoglobin. All five were women. · High ALT: 14.3% of patients (4/28) had high ALT. All four were aged 50+. · LDL–HDL: 58.3% of patients with high LDL (7/12) also had low HDL, compared with none of the 16 patients with within-range LDL. NB: These findings describe the synthetic course dataset and use the supplied exercise thresholds. They cannot be generalised to real patient populations.
Patient checker	The “Patient Checker” sheet allows users to enter results for one patient and immediately compare them with the supplied reference intervals. Enter values in the highlighted input cells, using the units shown. Each result is labelled Low, Within range or High. For HDL, only Low or Not low is reported because its upper reference limit remains unresolved. The checker is a demonstration tool for this project, not a diagnostic tool. Entered values do not update the study dataset, PivotTables or charts. Clear the input cells before checking another patient.